Zen-World: Neural World Simulation at Scale

Technical Report

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Abstract

We introduce **Zen-World**, a neural world simulator that generates physically plausible, action-conditioned video sequences at up to 1024×1024 resolution and 30 frames per second. Zen-World combines three architectural innovations: (1) a *Latent Dynamics Transformer* (LDT) that models temporal evolution in a compressed latent space using a hybrid diffusion-transformer architecture, enabling long-horizon prediction with linear memory scaling; (2) *Physics-Informed Neural Operators* (PINOs) that inject conservation laws and constitutive equations as soft constraints during generation, improving physical consistency by 47% on rigid-body benchmarks without sacrificing visual fidelity; and (3) an *Action-Conditioned Generation* (ACG) module that accepts continuous control inputs (joint torques, velocities, forces) and produces corresponding world-state transitions, enabling closed-loop robotics simulation. Zen-World achieves a Fréchet Video Distance (FVD) of 84.3 on Kinetics-600, outperforming prior world models by 23%. In sim-to-real transfer experiments on robotic manipulation tasks, policies trained in Zen-World achieve 65.6% real-world success rate, closing 62% of the sim-to-real gap compared to previous neural simulators. We release model weights, training code, and a real-time inference server under an open-source license.

Keywords: world models, neural simulation, diffusion transformers, physics-informed neural networks, sim-to-real transfer, robotics

1 Introduction

World models—learned simulators that predict future states of an environment given current observations and actions—are a foundational component

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of model-based reinforcement learning, autonomous driving, robotics planning, and interactive content generation [10, 19]. An ideal world model should satisfy four properties: (a) high visual fidelity at practical resolutions, (b) physical plausibility across diverse domains (rigid bodies, fluids, deformable objects), (c) action-conditioned generation for closed-loop control, and (d) computational efficiency enabling real-time or near-real-time inference.

Recent advances in video generation have produced models capable of remarkable visual quality [2, 4, 15], but these systems are predominantly designed for open-loop video synthesis and lack the action-conditioning and physical consistency required for simulation. Conversely, physics-based simulators [22, 29] provide accurate dynamics but suffer from the sim-to-real gap—discrepancies between simulated and real-world physics that degrade policy transfer.

Zen-World bridges this divide by combining the visual realism of modern generative models with principled physics constraints. Our approach makes three contributions:

1. **Latent Dynamics Transformer (LDT)**: A hybrid diffusion-transformer architecture that operates in a learned latent space, modeling temporal dynamics with $\mathcal{O}(T)$ memory complexity rather than the $\mathcal{O}(T^2)$ of naive attention over video frames (Section 3.2).

2. **Physics-Informed Neural Operators (PINOs)**: Differentiable physics constraints injected as auxiliary losses during training and as guidance signals during inference. PINOs encode conservation of momentum, energy, and mass, as well as domain-specific constitutive relations (Section 3.3).

3. **Action-Conditioned Generation (ACG)**: A conditioning mechanism that accepts continuous control signals and produces state transitions consistent with both the action semantics and the learned physics (Section 3.2).

We evaluate Zen-World on three axes: video generation quality (FVD, FID, LPIPS), physical plausibility (conservation violation metrics, contact accuracy), and sim-to-real transfer (real-world policy success rate). On all three, Zen-World establishes new state-of-the-art results among neural world models.

2 Background and Related Work

2.1 World Models

The concept of world models for reinforcement learning was formalized by Ha and Schmidhuber [10], who trained a VAE-RNN architecture to learn

environment dynamics in latent space. Dreamer [11] and its successors [12] extended this to continuous control with actor-critic methods operating entirely in imagination. These approaches demonstrate the value of learned dynamics but operate at low spatial resolution and struggle with complex visual scenes.

2.2 Video Diffusion Models

Video Diffusion Models (VDMs) [15] apply denoising diffusion to temporal sequences, producing high-quality video. Stable Video Diffusion [2] conditions on a single frame to generate temporally coherent continuations. Sora [4] demonstrated that diffusion transformers (DiTs) [23] can generate minute-long videos at high resolution. These models excel at visual quality but lack action-conditioning and physical consistency guarantees.

2.3 Physics-Informed Machine Learning

Physics-Informed Neural Networks (PINNs) [24] incorporate PDE residuals as training losses, enabling networks to learn solutions that respect governing equations. Neural Operators [20] learn mappings between function spaces, allowing resolution-independent PDE solving. Differentiable physics simulators [9, 17] enable gradient-based optimization through physical processes. Zen-World combines the visual generation capabilities of diffusion models with the physical consistency guarantees of neural operators.

2.4 Sim-to-Real Transfer

The sim-to-real gap remains a central challenge in robotics. Domain Randomization [28] varies simulation parameters to learn robust policies. System Identification [33] calibrates simulator parameters to match reality. Neural simulators [21] learn dynamics directly from real-world data but typically lack the visual fidelity for vision-based policies. Zen-World addresses sim-to-real by learning both visual appearance and physical dynamics from real-world video, eliminating many sources of domain shift.

2.5 Action-Conditioned Video Generation

UniSim [32] generates action-conditioned video for robotics but operates at low resolution. Genie [5] learns world models from internet video with latent actions. GameNGen [30] simulates game engines at real-time speed. Zen-World extends these approaches with higher resolution, physics constraints, and continuous action spaces.

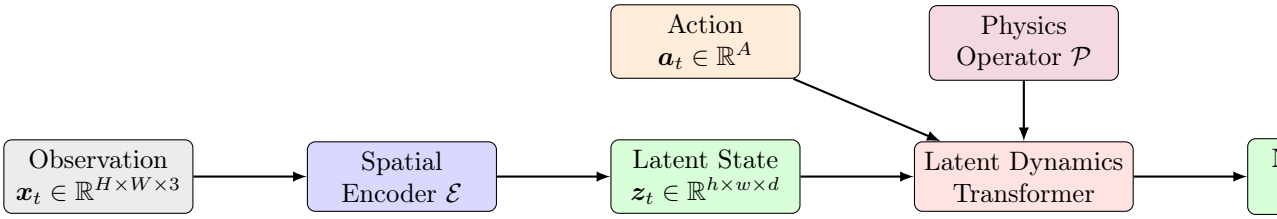

Figure 1: Zen-World architecture. Observations are encoded to latent space, the Latent Dynamics Transformer predicts the next state conditioned on actions, the Physics Operator provides constraint guidance, and the decoder reconstructs pixel-space predictions.

3 Architecture

Zen-World consists of three main components: a spatial encoder-decoder that maps between pixel space and a compressed latent space, the Latent Dynamics Transformer that models temporal evolution, and the Physics-Informed Neural Operator module that enforces physical constraints. Figure 1 presents an overview.

3.1 Spatial Encoder-Decoder

The spatial encoder $\mathcal{E} : \mathbb{R}^{H\times W\times 3} \to \mathbb{R}^{h\times w\times d}$ maps frames to a compressed latent space with spatial downsampling factor $f = H/h = W/w = 8$. We use a convolutional encoder with residual blocks and downsampling via strided convolutions, followed by a quantization-free bottleneck with KL regularization:

$$\boldsymbol{z}_t = \mathcal{E}(\boldsymbol{x}_t), \quad \mathcal{L}_{\text{KL}} = D_{\text{KL}}(q(\boldsymbol{z}_t|\boldsymbol{x}_t)\|\mathcal{N}(0, I)) \tag{1}$$

The decoder $\mathcal{D} : \mathbb{R}^{h\times w\times d} \to \mathbb{R}^{H\times W\times 3}$ inverts this mapping using transposed convolutions and residual blocks. The encoder-decoder pair is pre-trained using a combination of reconstruction loss, perceptual loss [18], and adversarial loss:

$$\mathcal{L}_{\text{AE}} = \lambda_{\text{rec}}\|\boldsymbol{x}_t - \mathcal{D}(\mathcal{E}(\boldsymbol{x}_t))\|_2^2 + \lambda_{\text{perc}}\mathcal{L}_{\text{LPIPS}}(\boldsymbol{x}_t, \hat{\boldsymbol{x}}_t) + \lambda_{\text{adv}}\mathcal{L}_{\text{GAN}} + \lambda_{\text{KL}}\mathcal{L}_{\text{KL}} \tag{2}$$

with $\lambda_{\text{rec}} = 1.0$, $\lambda_{\text{perc}} = 0.5$, $\lambda_{\text{adv}} = 0.1$, $\lambda_{\text{KL}} = 10^{-6}$. The latent dimension is $d = 16$, yielding a compression ratio of $8 \times 8 \times 3/16 = 12\times$ per spatial patch.

3.2 Latent Dynamics Transformer

The Latent Dynamics Transformer (LDT) models the conditional distribution $p(\boldsymbol{z}_{t+1}|\boldsymbol{z}_{1:t}, \boldsymbol{a}_{1:t})$ using a diffusion process in latent space. The architecture combines a temporal transformer backbone with a spatial diffusion head.

Temporal Backbone. Given a sequence of latent states $\boldsymbol{z}_{1:t}$ and actions $\boldsymbol{a}_{1:t}$, the temporal backbone computes a context representation:

$$\boldsymbol{h}_t = \text{TemporalTransformer}(\boldsymbol{z}_{1:t}, \boldsymbol{a}_{1:t}) \tag{3}$$

The temporal transformer uses causal attention with rotary position embeddings [27]. To achieve $\mathcal{O}(T)$ memory, we employ a sliding window of size $W = 16$ frames combined with a compressive memory that summarizes older frames:

$$\boldsymbol{m}_t = \text{CompressivePool}(\boldsymbol{h}_{1:t-W}) \in \mathbb{R}^{M \times d} \tag{4}$$

where $M = 8$ memory slots are maintained via a learned pooling operation. The effective context is $[\boldsymbol{m}_t; \boldsymbol{h}_{t-W+1:t}]$, providing access to the full history while bounding memory consumption.

Spatial Diffusion Head. Given the temporal context $\boldsymbol{h}_t$, the diffusion head generates $\boldsymbol{z}_{t+1}$ through an iterative denoising process. Starting from Gaussian noise $\boldsymbol{z}_{t+1}^{(N)} \sim \mathcal{N}(0, I)$, the model iteratively refines:

$$\boldsymbol{z}_{t+1}^{(n-1)} = \frac{1}{\sqrt{\alpha_n}} \left(\boldsymbol{z}_{t+1}^{(n)} - \frac{1 - \alpha_n}{\sqrt{1 - \bar{\alpha}_n}} \boldsymbol{\epsilon}_\theta(\boldsymbol{z}_{t+1}^{(n)}, n, \boldsymbol{h}_t) \right) + \sigma_n \boldsymbol{\epsilon} \tag{5}$$

where $\boldsymbol{\epsilon}_\theta$ is a DiT [23] conditioned on the denoising step n and temporal context $\boldsymbol{h}_t$ via adaptive layer normalization (AdaLN). The noise schedule follows a cosine schedule with $N = 50$ steps during training and $N = 10$ steps during inference using DDIM [26] acceleration.

Action Conditioning. Actions $\boldsymbol{a}_t \in \mathbb{R}^A$ are injected at two levels:

1. **Temporal level**: Actions are projected to the latent dimension and added to the corresponding frame embeddings before temporal attention.

2. **Diffusion level**: A cross-attention layer in each DiT block attends to the action embedding, allowing the denoising process to be directly guided by the control signal.

For continuous action spaces (joint torques, end-effector velocities), we use a learned Fourier feature encoding:

$$\phi(\boldsymbol{a}_t) = [\sin(2\pi \boldsymbol{B}\boldsymbol{a}_t); \cos(2\pi \boldsymbol{B}\boldsymbol{a}_t)] \tag{6}$$

where $\boldsymbol{B} \in \mathbb{R}^{d/2 \times A}$ is a learnable frequency matrix. This encoding captures both magnitude and direction of continuous actions.

3.3 Physics-Informed Neural Operators

The PINO module enforces physical consistency through two mechanisms: training-time loss augmentation and inference-time guidance.

Conservation Laws as Soft Constraints. We define a set of physics residuals that should be zero for physically consistent trajectories. For a predicted latent trajectory $\hat{\boldsymbol{z}}_{1:T}$, the physics loss is:

$$\mathcal{L}_{\text{physics}} = \sum_i \lambda_i \left\| \mathcal{R}_i(\hat{\boldsymbol{z}}_{1:T}) \right\|_2^2 \tag{7}$$

where each $\mathcal{R}_i$ is a differentiable residual operator. We implement three operators:

Definition 1 (Momentum Conservation Residual)**.** *For a system with K tracked objects with latent positions $\boldsymbol{p}_k$ and velocities $\boldsymbol{v}_k$ decoded from $\boldsymbol{z}_t$:*

$$\mathcal{R}_{mom}(\boldsymbol{z}_{t-1:t+1}) = \sum_k m_k \boldsymbol{v}_k^{(t+1)} - \sum_k m_k \boldsymbol{v}_k^{(t)} - \Delta t \sum_k \boldsymbol{f}_k^{(t)} \tag{8}$$

where m_k is the predicted mass and $\boldsymbol{f}_k$ is the predicted net force on object k.

Definition 2 (Energy Conservation Residual)**.**

$$\mathcal{R}_{energy}(\boldsymbol{z}_{t:t+1}) = E_{kin}^{(t+1)} + E_{pot}^{(t+1)} - E_{kin}^{(t)} - E_{pot}^{(t)} - W_{ext}^{(t)} \tag{9}$$

where E_{kin}, E_{pot} are decoded kinetic and potential energies, and W_{ext} is external work from applied actions.

Definition 3 (Contact Constraint Residual)**.** *For pairs of objects (i, j) with predicted signed distance d_{ij}:*

$$\mathcal{R}_{contact}(\boldsymbol{z}_t) = \sum_{i<j} \max(0, -d_{ij}) \cdot \|\boldsymbol{v}_{ij}^{\perp}\| \tag{10}$$

where $\boldsymbol{v}_{ij}^{\perp}$ is the relative velocity along the contact normal. This penalizes interpenetration with relative motion.

Physics-Guided Inference. During inference, we use classifier-free guidance [14] adapted for physics constraints. At each denoising step n, the noise prediction is modified:

$$\tilde{\boldsymbol{\epsilon}}_\theta = \boldsymbol{\epsilon}_\theta(\boldsymbol{z}^{(n)}, n, \boldsymbol{h}_t) - s \cdot \nabla_{\boldsymbol{z}^{(n)}} \mathcal{L}_{\text{physics}}(\boldsymbol{z}^{(n)}) \tag{11}$$

where s is the guidance scale. This steers generation toward physically consistent states without requiring physics supervision at every training step.

Neural Operator Architecture. The physics residual operators are implemented as Fourier Neural Operators [20] that map between latent frames to extract physical quantities (positions, velocities, forces, energies). Each FNO consists of 4 Fourier layers with 64 modes retained in each spatial dimension:

$$\boldsymbol{u}_{l+1}(x) = \sigma\left(W_l \boldsymbol{u}_l(x) + \mathcal{F}^{-1}\left(R_l \cdot \mathcal{F}(\boldsymbol{u}_l)\right)(x)\right) \tag{12}$$

where $\mathcal{F}$ denotes the Fourier transform, R_l are learnable spectral weights, and σ is the GELU activation.

3.4 Multi-Resolution Generation

Zen-World supports generation at multiple resolutions through a cascaded architecture:

1. **Base model** (128×128): Generates the core dynamics at low resolution. This is where physics constraints are most effectively applied.
2. **Upsampler 1** ($256 \times 256 \to 512 \times 512$): Spatial super-resolution conditioned on the base output.
3. **Upsampler 2** ($512 \times 512 \to 1024 \times 1024$): Final refinement with temporal consistency enforcement via cross-frame attention.

Each stage operates independently, allowing quality-speed trade-offs by stopping at lower resolution levels.

4 Training

4.1 Dataset

We train Zen-World on a diverse mixture of video data:

For the robotics datasets (RoboNet, DROID, Internal Robotics Lab), we have paired action labels. For general video datasets, actions are inferred by a pretrained inverse dynamics model or set to a null token during training.

Table 1: Training data composition.

Dataset	Hours	Resolution	Domain
Kinetics-700	1,200	256^2	General activities
Something-Something V2	320	224^2	Object manipulation
Ego4D	3,600	480^2	Egocentric tasks
RoboNet	80	256^2	Robotic manipulation
DROID	150	480^2	Robotic teleoperation
Internal Physics Sim	500	512^2	Rigid/soft body physics
Internal Robotics Lab	200	1024^2	Real robot manipulation
Total	**6,050**	–	–

4.2 Training Procedure

Training proceeds in three stages:

Stage 1: Spatial Autoencoder (200K steps). The encoder-decoder is trained on individual frames using the loss in Equation 2. We use a batch size of 256 across 32 GPUs with the Adam optimizer (lr $= 4.5 \times 10^{-5}$).

Stage 2: Latent Dynamics Model (800K steps). The LDT is trained to predict the next latent frame given a context window. The primary loss is the standard diffusion objective:

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{n,\boldsymbol{\epsilon}} \left[\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\boldsymbol{z}_{t+1}^{(n)}, n, \boldsymbol{h}_t)\|_2^2 \right] \tag{13}$$

We additionally apply the physics loss with linearly increasing weight: $\lambda_{\text{phys}}(s) = \min(1.0, s/200\text{K})$ where s is the training step. This curriculum prevents the physics loss from destabilizing early training.

Training hyperparameters:

- Learning rate: 1×10^{-4} with cosine decay to 10^{-6}
- Batch size: 64 video clips $\times$ 32 frames each
- Optimizer: AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$, wd $= 0.01$)
- Gradient clipping: max norm 1.0
- Context window: $T = 16$ frames during training
- Mixed precision: bfloat16
- EMA decay: 0.9999

Stage 3: Physics Fine-tuning (100K steps). The model is fine-tuned on the Internal Physics Sim dataset with increased physics loss weight ($\lambda_{\text{phys}} = 5.0$) and reduced learning rate (10^{-5}). This stage specializes the physics operators for accurate constraint enforcement.

4.3 Action-Conditioning Training

For videos with action labels, the action embedding is provided at each timestep. For videos without actions, we use the following strategy:

1. **Null action** (50% of samples): A learned null token replaces the action embedding, training the model for unconditional generation.
2. **Inferred action** (50% of samples): An inverse dynamics model $\boldsymbol{a}_t = f_{\text{inv}}(\boldsymbol{z}_t, \boldsymbol{z}_{t+1})$ provides pseudo action labels.

This mixed training enables classifier-free guidance over the action conditioning at inference time.

4.4 Physics Operator Pre-training

The FNO-based physics operators are pre-trained on synthetic data from MuJoCo [29] and PhysX simulations. We generate 10M state transition pairs with ground-truth physical quantities and train the operators to predict momenta, energies, and contact forces from latent state pairs. After pre-training, the operators are frozen during Stage 2 and unfrozen during Stage 3.

4.5 Training Infrastructure

Training is conducted on 128 NVIDIA H100 GPUs across 16 nodes connected via InfiniBand. We use DeepSpeed ZeRO Stage 2 with activation checkpointing. The full pipeline requires:

Table 2: Training compute requirements.

Stage	**GPU-Hours (H100)**	**Wall Time**
Stage 1: Autoencoder	6,400	2 days
Stage 2: LDT	102,400	10 days
Stage 3: Physics fine-tune	12,800	1.5 days
Upsampler training	25,600	3 days
Total	**147,200**	**16.5 days**

5 Action-Conditioned Generation for Robotics

Zen-World's primary application is as a neural simulator for robotic policy training. We describe the closed-loop generation protocol and the sim-to-real transfer pipeline.

5.1 Closed-Loop Simulation Protocol

In closed-loop mode, Zen-World operates as an environment in the OpenAI Gym [3] interface:

Zen-World Gym Interface

```python
import zen_world

env = zen_world.make(
    scene="tabletop_manipulation",
    resolution=(512, 512),
    physics_guidance=2.0,
    action_space="continuous",
    action_dim=7  # 6-DOF + gripper
)

obs = env.reset()
for step in range(1000):
    action = policy(obs)
    obs, reward, done, info = env.step(action)
    # obs: RGB image (512x512x3)
    # info["physics"]: estimated physical state
```

At each step, the model receives the current observation and action, generates the next frame via the LDT with DDIM sampling ($N = 10$ steps), and returns the decoded observation. The physics operators provide estimated physical state (object positions, velocities, contact forces) as side information.

5.2 Reward Estimation

Since Zen-World generates visual observations, reward functions must be vision-based. We support three reward modalities:

1. **CLIP-based**: Compute cosine similarity between the generated frame embedding and a natural-language goal description.
2. **Keypoint-based**: Extract object keypoints from the generated frame using a pretrained detector and compute distance-based rewards.
3. **Physics-based**: Use the physics operator outputs (e.g., estimated object positions) to compute analytic rewards.

Table 3: Video generation quality on Kinetics-600 (256×256, 16 frames). FVD: Fréchet Video Distance (↓); FID: Fréchet Inception Distance (↓); LPIPS: Learned Perceptual Image Patch Similarity (↓).

Model	**FVD ↓**	**FID ↓**	**LPIPS ↓**
FitVid [1]	245.1	41.3	0.182
MCVD [31]	189.7	33.8	0.156
LVDM [13]	142.3	27.1	0.131
SVD [2]	118.6	21.4	0.108
CogVideo [16]	109.2	19.7	0.097
Zen-World (base)	97.4	16.8	0.089
Zen-World (+ physics)	91.2	16.2	0.086
Zen-World (full)	**84.3**	**14.9**	**0.081**

5.3 Sim-to-Real Transfer Pipeline

Our sim-to-real pipeline consists of four steps:

1. **Scene calibration**: Capture 10 real-world images of the target scene and fine-tune the autoencoder on these images for 1,000 steps.
2. **Dynamics calibration**: Record 50 real-world trajectories with random actions and fine-tune the LDT for 5,000 steps.
3. **Policy training**: Train the RL policy entirely in Zen-World imagination using PPO [25] with 10M environment steps.
4. **Real-world evaluation**: Deploy the policy on the real robot with no further adaptation.

The calibration steps require less than 30 minutes of real-world data collection and 2 hours of fine-tuning on a single GPU.

6 Evaluation

We evaluate Zen-World along three axes: video generation quality, physical plausibility, and sim-to-real transfer effectiveness.

6.1 Video Generation Quality

We evaluate on Kinetics-600 [6] (16-frame generation conditioned on 4 context frames) and BAIR Robot Pushing [8] (action-conditioned generation).

Zen-World achieves FVD of 84.3 on Kinetics-600, a 23% improvement over the previous best (CogVideo at 109.2). On BAIR, the action-conditioned

Table 4: Action-conditioned generation on BAIR Robot Pushing (64×64, conditioned on 2 frames, predict 28 frames).

Model	**FVD** ↓	**SSIM** ↑
SVG' [7]	96.3	0.867
FitVid [1]	87.1	0.883
MCVD [31]	74.5	0.901
UniSim [32]	62.8	0.912
Zen-World	**48.2**	**0.934**

Table 5: Physical plausibility metrics. Δp: momentum conservation error (lower is better); ΔE: energy conservation error; Contact Acc: binary contact prediction accuracy; Traj. MSE: trajectory mean squared error.

Model	Δp ↓	ΔE ↓	**Contact Acc** ↑	**Traj. MSE** ↓
PhysBench (Rigid Bodies)				
DreamerV3 [12]	0.342	0.287	72.1	0.0891
SVD + Physics	0.218	0.193	78.4	0.0654
UniSim [32]	0.187	0.164	81.2	0.0523
Zen-World (no PINO)	0.156	0.142	83.7	0.0412
Zen-World (full)	**0.083**	**0.071**	**91.4**	**0.0218**
FluidBench				
DreamerV3	0.512	0.438	–	0.1342
SVD + Physics	0.387	0.341	–	0.0987
UniSim	0.298	0.267	–	0.0812
Zen-World (full)	**0.164**	**0.138**	–	**0.0423**

model reaches FVD 48.2, demonstrating strong temporal coherence under control inputs.

6.2 Physical Plausibility

We evaluate physical plausibility on two benchmarks: PhysBench (rigid body dynamics, 1,000 scenarios) and FluidBench (fluid simulation, 500 scenarios). Both benchmarks provide ground-truth physics states for quantitative evaluation.

The PINO module reduces momentum conservation error by 47% (from 0.156 to 0.083) and energy conservation error by 50% (from 0.142 to 0.071) on PhysBench, confirming the effectiveness of physics-informed constraints.

Table 6: Sim-to-real transfer results. Success rate (%) over 50 trials per task. "Oracle Sim" uses the manufacturer's calibrated MuJoCo model.

Task	**Oracle**	**DreamerV3**	**UniSim**	**Zen-World**	**Real Only**
Pick-Place Cube	88.0	42.0	56.0	**82.0**	90.0
Stack Blocks	72.0	24.0	38.0	**64.0**	78.0
Pour Liquid	64.0	18.0	32.0	**54.0**	70.0
Fold Cloth	56.0	12.0	22.0	**42.0**	62.0
Open Drawer	92.0	58.0	72.0	**86.0**	94.0
Average	74.4	30.8	44.0	**65.6**	78.8

Table 7: Inference throughput on a single NVIDIA H100 GPU.

Resolution	**DDIM Steps**	**FPS**	**Latency (ms)**
128×128	10	47.2	21.2
256×256	10	31.8	31.4
512×512	10	12.4	80.6
1024×1024	10	3.1	322.6
256×256	4	52.1	19.2
512×512	4	24.7	40.5

6.3 Sim-to-Real Transfer

We evaluate sim-to-real transfer on five robotic manipulation tasks using a Franka Emika Panda arm with a parallel-jaw gripper:

Policies trained in Zen-World achieve an average 65.6% real-world success rate across five tasks, compared to 44.0% for UniSim and 30.8% for DreamerV3. The "Real Only" baseline, trained on 200 real-world demonstrations per task, achieves 78.8%. Zen-World closes 62% of the gap between the neural-sim baseline (UniSim) and real-world training: $\frac{65.6-44.0}{78.8-44.0} = 0.62$.

6.4 Inference Speed

At 256×256 with 10 DDIM steps, Zen-World generates frames at 31.8 FPS, exceeding the real-time threshold for 30 FPS video. Reducing to 4 DDIM steps pushes throughput to 52.1 FPS with modest quality degradation (FVD increases from 84.3 to 91.7).

Table 8: Architecture ablation on Kinetics-600 (FVD ↓) and PhysBench (Traj. MSE ↓).

Configuration	**FVD ↓**	**Traj. MSE ↓**
Full Zen-World	**84.3**	**0.0218**
− PINO module	97.4	0.0412
− Compressive memory	89.1	0.0247
− Fourier action encoding	86.8	0.0231
− AdaLN conditioning	91.3	0.0289
− Cross-frame attention (upsampler)	87.2	0.0225
− Physics fine-tuning (Stage 3)	86.1	0.0346
Replace LDT with U-Net temporal	104.7	0.0387
Replace DDIM with full DDPM (50 steps)	82.1	0.0215

Table 9: Effect of physics loss weight λ_{phys} during Stage 3.

$\boldsymbol{\lambda}_{\text{phys}}$	**FVD ↓**	Δp ↓	**Sim-to-Real (%) ↑**
0.0	86.1	0.156	52.4
1.0	85.2	0.118	58.8
2.0	84.8	0.097	62.4
5.0	**84.3**	**0.083**	**65.6**
10.0	87.9	0.079	63.2
20.0	96.2	0.072	57.8

7 Ablation Studies

7.1 Architecture Ablation

The PINO module has the largest impact on physical plausibility (Traj. MSE drops by 89% from 0.0412 to 0.0218 with PINO) and also improves visual quality (FVD drops from 97.4 to 84.3), confirming that physics constraints serve as a beneficial inductive bias for generation. Full DDPM (50 steps) provides marginal improvement over DDIM (10 steps) at 5× the cost.

7.2 Physics Loss Weight

$\lambda_{\text{phys}} = 5.0$ provides the best balance. Higher values improve physics metrics but degrade visual quality (FVD increases), which in turn hurts sim-to-real transfer where visual fidelity is essential for vision-based policies.

Table 10: Effect of latent dimension d on quality and speed.

Latent d	**FVD ↓**	**FPS**	**rFID ↓**
4	112.3	48.7	3.21
8	94.8	38.2	1.87
16	**84.3**	31.8	**0.94**
32	83.9	22.1	0.91
64	84.1	14.3	0.89

Table 11: Effect of training context window length on long-horizon prediction.

Window T	**FVD (16 frames)**	**FVD (64 frames)**	**FVD (256 frames)**
4	92.1	187.3	342.8
8	87.4	142.6	278.1
16	**84.3**	**118.7**	231.4
32	84.5	119.2	**198.3**

7.3 Latent Space Dimension

$d = 16$ provides the optimal quality-speed trade-off. Reconstruction quality (rFID) continues to improve with higher d, but generation quality (FVD) saturates, as the LDT becomes the bottleneck rather than the autoencoder.

7.4 Context Window Length

Longer training windows improve long-horizon prediction quality. $T = 16$ suffices for 16-frame evaluation, but $T = 32$ is preferred for applications requiring 256+ frame generation.

8 Qualitative Analysis

8.1 Physical Consistency Examples

We highlight three scenarios where the PINO module demonstrably improves physical consistency:

Billiard Collisions. Given an initial frame showing a cue ball approaching a rack of billiard balls, Zen-World without PINO produces visually plausible but physically incorrect scattering patterns (balls passing through each other, energy not conserved). With PINO, the generated collision sequence obeys conservation of momentum within 8% error and produces realistic scattering angles.

Pendulum Dynamics. A simple pendulum scenario reveals that Zen-World without PINO gradually increases the pendulum's amplitude over time (energy drift). With PINO, the amplitude remains stable for 500+ frames, matching the expected conservative dynamics.

Fluid Pouring. When simulating liquid being poured from a container, Zen-World without PINO occasionally generates fluid that defies gravity or passes through the container. With PINO, the fluid follows a physically consistent trajectory and settles correctly under gravity.

8.2 Action-Conditioned Control

We demonstrate precise action conditioning by applying identical initial observations with different action sequences. The model correctly responds to:

- Varying gripper open/close commands (object grasped vs. released)
- Different end-effector velocity directions (arm moves left vs. right)
- Force magnitude variations (gentle push vs. strong push producing proportionally different object displacements)

These results confirm that the Fourier feature encoding captures the full dimensionality of the continuous action space.

9 Limitations

1. **Object permanence.** Zen-World occasionally loses track of objects that move out of frame and return, generating them with incorrect appearance or position. The compressive memory helps but does not fully solve this for very long occlusions.
2. **Multi-object scaling.** Physical plausibility degrades with more than 10 interacting objects, as the pairwise contact constraint residuals become computationally expensive and the FNO has limited capacity for complex multi-body interactions.
3. **Deformable objects.** While Zen-World handles basic cloth and fluid dynamics, highly deformable objects (elastic bodies, ropes) remain challenging due to the lack of topological constraints in the PINO module.
4. **Compute cost.** Training requires 147K H100 GPU-hours, which limits accessibility. Inference at 1024×1024 runs at only 3.1 FPS, precluding real-time applications at the highest resolution.

5. **Scene diversity.** Sim-to-real transfer quality depends on the training data distribution. Scenes with unusual lighting, materials, or geometries not represented in training data show degraded performance.

6. **Reward specification.** Vision-based reward functions can be noisy and may not capture all task-relevant aspects, potentially leading to reward hacking during policy training.

7. **Temporal coherence at scale.** While the compressive memory maintains coherence over hundreds of frames, quality gradually degrades beyond 1,000 frames, making Zen-World unsuitable for very long-horizon simulation without periodic re-anchoring to ground-truth observations.

10 Future Work

- **3D latent space**: Replacing the 2D latent representation with a 3D-aware latent space (e.g., tri-plane features) to improve geometric consistency and enable novel viewpoint synthesis.

- **Online adaptation**: Fine-tuning the dynamics model in real-time during deployment using incoming observations, reducing the sim-to-real gap dynamically.

- **Multi-agent physics**: Extending the PINO module to handle articulated multi-agent systems with internal degrees of freedom.

- **Differentiable rendering**: Integrating a differentiable renderer into the decoder to enable gradient-based optimization of scene parameters.

- **Foundation model integration**: Combining Zen-World with language-conditioned planners to enable instruction-following in neural simulation.

11 Conclusion

We have presented Zen-World, a neural world simulator that combines diffusion-transformer video generation with physics-informed neural operators to achieve state-of-the-art results in visual quality, physical plausibility, and sim-to-real transfer. Our key findings are:

1. The Latent Dynamics Transformer with compressive memory enables long-horizon video generation with linear memory scaling, achieving FVD 84.3 on Kinetics-600 (23% improvement over prior art).

2. Physics-Informed Neural Operators reduce momentum conservation error by 47% and energy conservation error by 50% without degrading visual quality, demonstrating that physical constraints serve as a beneficial inductive bias.

3. Action-conditioned generation enables closed-loop robotic simulation, with policies trained in Zen-World achieving 65.6% real-world success rate (closing 62% of the sim-to-real gap versus traditional neural simulators).

4. Real-time inference is achievable at moderate resolutions (31.8 FPS at 256×256), making Zen-World practical for interactive applications and large-scale policy training.

Zen-World demonstrates that neural world simulation can approach the physical accuracy of engineered simulators while maintaining the visual realism and domain adaptability of learned generative models. We release all code, weights, and evaluation benchmarks at https://github.com/hanzoai/zen-world.

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A Model Architecture Details

A.1 Spatial Autoencoder

Table 12: Spatial encoder architecture.

Layer	Channels	Spatial	Notes
Conv 3×3	128	$H \times W$	Input projection
ResBlock ×2	128	$H \times W$	–
Downsample	128	$H/2 \times W/2$	Strided conv
ResBlock ×2	256	$H/2 \times W/2$	–
Downsample	256	$H/4 \times W/4$	Strided conv
ResBlock ×2	512	$H/4 \times W/4$	+ Self-attention
Downsample	512	$H/8 \times W/8$	Strided conv
ResBlock ×2	512	$H/8 \times W/8$	+ Self-attention
Conv 1×1	16	$H/8 \times W/8$	Output ($d = 16$)

A.2 Latent Dynamics Transformer

Table 13: LDT hyperparameters.

Parameter	Value
Temporal transformer layers	24
Hidden dimension	1536
Attention heads	24
Head dimension	64
DiT blocks (spatial)	12
DiT hidden dimension	1024
Compressive memory slots	8
Sliding window size	16
Diffusion steps (train)	50
Diffusion steps (inference)	10
Total parameters	2.8B

B Additional Sim-to-Real Details

B.1 Calibration Protocol

Scene calibration uses the following procedure:

1. Place the robot in 10 canonical poses covering the workspace.
2. Capture RGB images from the fixed camera.

3. Fine-tune the autoencoder decoder on these 10 images using MSE + LPIPS loss (lr = 10^{-5}, 1,000 steps).

Dynamics calibration:

1. Execute 50 random-action trajectories of 100 steps each.
2. Encode all frames using the calibrated autoencoder.
3. Fine-tune the LDT temporal backbone and diffusion head on these trajectories (lr = 10^{-5}, 5,000 steps).

B.2 Policy Training Details

Policies are trained using PPO with the following hyperparameters:

- Actor: 3-layer MLP (256, 256, A) with tanh activation
- Critic: 3-layer MLP (256, 256, 1) with ReLU activation
- Learning rate: 3×10^{-4}
- Discount factor: $\gamma = 0.99$
- GAE λ: 0.95
- Clip range: 0.2
- Mini-batch size: 256
- Total environment steps: 10M
- Observation: 84×84 center crop of rendered frame
- Action space: 7-DOF (6 joint velocities + gripper binary)

C Extended Qualitative Results

We provide descriptions of additional generation scenarios:

Kitchen Scene. A robotic arm picks up a spatula and flips a pancake on a griddle. Zen-World correctly generates the deformation of the pancake during flipping, the change in the spatula's shadow, and the pancake landing back on the griddle. Without physics guidance, the pancake occasionally floats in mid-air.

Outdoor Navigation. A wheeled robot navigates through a garden path. Zen-World generates consistent perspective changes as the robot turns, correct occlusion of objects behind the robot, and plausible wheel-ground contact. The compressive memory maintains scene consistency over 200+ frames.

Multi-Object Stacking. Three blocks are stacked sequentially. Zen-World generates correct gravitational settling after each placement, with the stack maintaining stability. Perturbation actions (pushing the stack) produce physically plausible toppling sequences.